Analytic Stress in DFT+eDMFT with LAPW

Implemented Equations, Bird's-Eye Derivation, and the Direct-$C_{\mathbf{k}}$ Diagnostic

A self-contained technical working note

This note records the equations presently implemented in `dmft2`, their relationship to the stationary DFT+eDMFT Klein functional and to the WIEN2k LAPW stress kernels, and the remaining diagnostic needed when the projector metric is singular at individual $\mathbf{k}$ points. Every symbol is defined, and implemented equations are separated from exact target identities and proposed tests. The emphasis is on the generalized off-diagonal kinetic, incomplete-basis, and overlap matrices required by an interacting density matrix.

5 November 2026

Based on the DFT+eDMFT force formalism, the Blaha–Belbase–Tröster LAPW stress derivation, and the current `eDMFT_new/src/dmft2` implementation.

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1 Purpose, scope, and status labels

The goal is to compute the first derivative of the stationary DFT+eDMFT thermodynamic functional under a homogeneous strain. The derivation must not differentiate the nonstationary total-energy expression. It must instead differentiate the Klein functional at its stationary point. This removes the implicit response of G and Σ—and hence the two-particle vertex—but retains every *explicit* dependence on the LAPW basis, its metric, and the normalized correlated projector.

Four kinds of statement are distinguished throughout:

Stationary identity	An exact one-particle identity following from stationarity and Dyson's equation.
Implemented	Present in the inspected `dmft2` source. It may still inherit the frozen-augmentation and finite-basis approximations of the underlying LAPW stress.
Proposed diagnostic	A controlled comparison that has not yet been added to the source.
Open audit item	A difference between the exact target and the present implementation that still requires numerical closure.

The note uses a fixed band-index DMFT window. The **k**-point weights are constant under strain. Muffin-tin radii, radial projector files, and the outer radius of projector 5 are held fixed in bohr. The base implementation uses fixed local correlated axes. A co-moving-axis correction is a later extension for a self-energy constrained to remain diagonal in a local frame.

2 Notation and dimensions

2.1 Indices and spaces

$\mathbf{k}$	Brillouin-zone point; $w_{\mathbf{k}}$ is its fixed normalized integration weight.
n, m	Kohn–Sham band indices inside the fixed DMFT window $W_{\mathbf{k}}$.
a, b	Bands stored by LAPW1 outside or inside $W_{\mathbf{k}}$; $a \notin W_{\mathbf{k}}$ denotes a complementary band.
ℓ, ℓ'	Composite correlated-orbital indices, including atom, shell, orbital, and spin block as needed.
ν	One of the six independent symmetric strain channels, ordered as (xx, yy, zz, yz, xz, xy).
α, β	Cartesian indices in $\{x, y, z\}$; $\nu \equiv (\alpha\beta)$.
$i\omega_j$	Fermionic Matsubara frequency, $\omega_j = (2j+1)\pi/\beta$.
β	Inverse temperature; the code uses Rydberg energy units.
Ω	Unstrained unit-cell volume.

Let $N_b(\mathbf{k})$ be the number of bands in the DMFT window and N_c the number of active correlated orbitals. The central matrices have dimensions

$$\rho_{\mathbf{k}}, \rho_{E,\mathbf{k}}, \epsilon_{\mathbf{k}}, G_{\mathbf{k}} \in \mathbb{C}^{N_b \times N_b}, \tag{2.1}$$

$$P_{\mathbf{k}}, U_{\mathbf{k}} \in \mathbb{C}^{N_b \times N_c}, \tag{2.2}$$

$$\widetilde{\Sigma}, S, X \in \mathbb{C}^{N_c \times N_c}, \tag{2.3}$$

$$C_{\mathbf{k}} \in \mathbb{C}^{N_c \times N_b}. \tag{2.4}$$

The trace convention is explicit. For example,

$$\mathrm{Tr}_c(C_{\mathbf{k}}U_{\mathbf{k},\nu}) = \sum_{\ell n}(C_{\mathbf{k}})_{\ell n}(U_{\mathbf{k},\nu})_{n\ell}. \tag{2.5}$$

No dagger belongs in Eq. (2.5); C is the rectangular one-particle gradient with the orientation chosen in Sec. 9.

2.2 Strain and stress convention

With deformation gradient

$$F = I + \varepsilon, \qquad \mathbf{r}' = F\mathbf{r}, \qquad \mathbf{G}' = F^{-T}\mathbf{G}, \tag{2.6}$$

the thermodynamic stress convention used in this note is

$$\sigma_{\alpha\beta} = \frac{1}{\Omega}\left.\frac{\partial\Gamma}{\partial\varepsilon_{\alpha\beta}}\right|_0, \qquad P = -\frac{1}{3}\,\mathrm{Tr}\,\sigma. \tag{2.7}$$

For an independent symmetric shear variable,

$$\frac{\partial\varepsilon_{\gamma\delta}}{\partial\varepsilon_{\alpha\beta}} = \frac{1}{2}(\delta_{\gamma\alpha}\delta_{\delta\beta} + \delta_{\gamma\beta}\delta_{\delta\alpha}). \tag{2.8}$$

Consequently, real-space and reciprocal-space vectors obey

$$\partial_{\alpha\beta}V_\gamma = \pm\frac{1}{2}(\delta_{\gamma\alpha}V_\beta + \delta_{\gamma\beta}V_\alpha), \tag{2.9}$$

with + for real-space vectors and − for reciprocal vectors. A shear finite difference must use the same convention or it will differ by a factor of two.

3 Bird's-eye stationary-functional derivation

3.1 Why the Klein functional is differentiated

Write the stationary DFT+eDMFT functional schematically as

$$\Gamma = \Gamma_{\mathrm{DFT}}[\rho] + \Gamma_{\mathrm{band}}[G; H, O, \Sigma^{\mathrm{emb}}] + \sum_R \left(\Phi_{\mathrm{imp},R}[G_R] - \Phi_{\mathrm{DC},R}[G_R]\right). \tag{3.1}$$

The detailed representation of individual double-counting and impurity terms is not needed for the first geometric derivative. At self-consistency,

$$\frac{\delta\Gamma}{\delta G} = 0, \qquad \frac{\delta\Gamma}{\delta\Sigma} = 0, \qquad \frac{\delta\Gamma}{\delta\rho} = 0. \tag{3.2}$$

Therefore the total derivative with respect to a strain channel ν reduces to the explicit derivative,

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}\varepsilon_\nu} = \left.\frac{\partial\Gamma}{\partial\varepsilon_\nu}\right|_{G,\Sigma,\rho}. \tag{3.3}$$

Terms containing $G_{,\nu}$, $\Sigma_{,\nu}$, and the impurity vertex cancel. This is a consequence of stationarity, not a frozen-self-energy approximation. Explicit changes of H, O, and the embedding projector do not cancel.

3.2 Nonorthogonal LAPW form

In an arbitrary nonorthogonal LAPW coefficient basis $\{\chi_K\}$,

$$G_\chi^{-1}(i\omega) = (i\omega + \mu)O - H_0 - \Sigma_\chi^{\mathrm{emb}}(i\omega). \tag{3.4}$$

Define the frequency-summed coefficient-space matrices

$$\rho_\chi = T\sum_j G_\chi(i\omega_j)e^{i\omega_j 0^+}, \tag{3.5}$$

$$\rho_{E,\chi} = T\sum_j (i\omega_j + \mu)G_\chi(i\omega_j)e^{i\omega_j 0^+}. \tag{3.6}$$

The stationary one-particle derivative is

$$D_\nu^{1p} = \sum_{\mathbf{k}} \mathrm{Re}\,\mathrm{Tr}\left[\rho_{\chi,\mathbf{k}}H_{\mathbf{k},\nu} - \rho_{E,\chi,\mathbf{k}}O_{\mathbf{k},\nu}\right] + D_\nu^{\mathrm{emb}}. \tag{3.7}$$

The remaining DFT density, Hartree, exchange-correlation, core, nuclear, and electrostatic terms assemble exactly as in the validated DFT stress. The new term D_ν^{emb} is the explicit derivative of the correlated embedding at fixed $\widetilde{\Sigma}$.

Stationary identity. Equation (3.7) is the organizing equation. It contains a paired $H_{,\nu}$ and $O_{,\nu}$ contraction. Their individual magnitudes can be very large, so numerical or analytic replacements must preserve the stationary pair.

4 Green functions, density matrices, and energy moments

In the orthonormal Kohn–Sham band basis of the DMFT window,

$$\Sigma_{\mathbf{k}}^{\mathrm{emb}}(i\omega) = U_{\mathbf{k}}\widetilde{\Sigma}(i\omega)U_{\mathbf{k}}^\dagger, \tag{4.1}$$

$$G_{\mathbf{k}}^{-1}(i\omega) = (i\omega + \mu)I - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}^{\mathrm{emb}}(i\omega), \tag{4.2}$$

where $\widetilde{\Sigma} = \Sigma_{\mathrm{imp}} - V_{\mathrm{DC}}$ and $\epsilon_{\mathbf{k}}$ is diagonal. The code constructs Hermitian frequency-summed matrices

$$\rho_{\mathbf{k}} = T\sum_j G_{\mathbf{k}}(i\omega_j)e^{i\omega_j 0^+}, \tag{4.3}$$

$$\rho_{E,\mathbf{k}} = T\sum_j (i\omega_j + \mu)G_{\mathbf{k}}(i\omega_j)e^{i\omega_j 0^+}. \tag{4.4}$$

Dyson's equation gives the left and right identities

$$B_{L,\mathbf{k}} \equiv \rho_{E,\mathbf{k}} - \epsilon_{\mathbf{k}}\rho_{\mathbf{k}} = T\sum_j \Sigma^{\mathrm{emb}}_{\mathbf{k}}(i\omega_j)G_{\mathbf{k}}(i\omega_j), \tag{4.5}$$

$$B_{R,\mathbf{k}} \equiv \rho_{E,\mathbf{k}} - \rho_{\mathbf{k}}\epsilon_{\mathbf{k}} = B^{\dagger}_{L,\mathbf{k}}. \tag{4.6}$$

The Hermitian self-energy moment is

$$M_{\Sigma,\mathbf{k}} \equiv \rho_{E,\mathbf{k}} - \frac{1}{2}\{\epsilon_{\mathbf{k}}, \rho_{\mathbf{k}}\} = \frac{1}{2}(B_{L,\mathbf{k}} + B_{R,\mathbf{k}}). \tag{4.7}$$

Using Eq. (4.7), the one-particle LAPW derivative can be separated into a DFT-like part and the genuinely new energy-moment metric part,

$$D^{1p}_{\nu} = D^{\mathrm{DFT-like}}_{\nu} - \sum_{\mathbf{k}} \mathrm{Re}\,\mathrm{Tr}[M_{\Sigma,\mathbf{k}}O_{\mathbf{k},\nu}], \tag{4.8}$$

$$D^{\mathrm{DFT-like}}_{\nu} = \sum_{\mathbf{k}} \mathrm{Re}\,\mathrm{Tr}\left[\rho_{\mathbf{k}}H_{\mathbf{k},\nu} - \frac{1}{2}\{\epsilon_{\mathbf{k}}, \rho_{\mathbf{k}}\}O_{\mathbf{k},\nu}\right]. \tag{4.9}$$

Important qualification. The $M_{\Sigma}O_{,\nu}$ term requires the complete generalized APW+LO metric derivative. The current implementation obtains an overlap kernel from a selected rearranged valence-correction kernel. Direct matrix tests show that this isolated analytic channel does not yet reproduce the full metric derivative in a material-independent way.

5 Generalized LAPW strain matrices

5.1 Fixed-coefficient basis response

Let

$$|\psi_{n\mathbf{k}}\rangle = \sum_a c_{an\mathbf{k}}|\phi_{a\mathbf{k}}\rangle \tag{5.1}$$

be a Kohn–Sham state expanded in the LAPW+LO basis. Its fixed-coefficient basis response is

$$|\psi^{B,\nu}_{n\mathbf{k}}\rangle = \sum_a c_{an\mathbf{k}}|\partial_\nu\phi_{a\mathbf{k}}\rangle. \tag{5.2}$$

The superscript B means "basis response": the eigenvector coefficients $c_{an\mathbf{k}}$ are held fixed. Eigenvector rotation into complementary bands is treated separately in Sec. 11.

Define one-sided matrices

$$b^{\nu}_{mn}(\mathbf{k}) = \langle\psi^{B,\nu}_{m\mathbf{k}}|H|\psi_{n\mathbf{k}}\rangle, \tag{5.3}$$

$$o^{\nu}_{mn}(\mathbf{k}) = \langle\psi^{B,\nu}_{m\mathbf{k}}|\psi_{n\mathbf{k}}\rangle. \tag{5.4}$$

Their physical Hermitian polarizations are

$$\mathcal{B}^{\nu} = b^{\nu} + b^{\nu\dagger}, \tag{5.5}$$

$$\mathcal{O}^{\nu} = o^{\nu} + o^{\nu\dagger}. \tag{5.6}$$

Thus

$$\mathcal{B}^{\nu}_{mn} = \langle \psi^{B,\nu}_{m} | H | \psi_n \rangle + \langle \psi_m | H | \psi^{B,\nu}_{n} \rangle, \tag{5.7}$$
$$\mathcal{O}^{\nu}_{mn} = \langle \psi^{B,\nu}_{m} | \psi_n \rangle + \langle \psi_m | \psi^{B,\nu}_{n} \rangle. \tag{5.8}$$

For $m = n$,

$$\mathcal{B}^{\nu}_{nn} - \epsilon_n \mathcal{O}^{\nu}_{nn} = 2\,\mathrm{Re}\langle \psi^{B,\nu}_{n} | (H - \epsilon_n) | \psi_n \rangle, \tag{5.9}$$

which is the diagonal expression in Blaha et al., Eq. (20). Blaha does not write the full $m \neq n$ matrices because ordinary DFT occupations are diagonal in the Kohn–Sham basis.

5.2 Kinetic strain matrix

For $\nu = (\alpha\beta)$, the generalized valence kinetic matrix is

$$T^{\alpha\beta}_{mn}(\mathbf{k}) = \frac{1}{2} \int_{\Omega} \mathrm{d}^3 r\, \psi^{*}_{m\mathbf{k}}(\mathbf{r}) (\partial_\alpha \partial_\beta + \partial_\beta \partial_\alpha) \psi_{n\mathbf{k}}(\mathbf{r}). \tag{5.10}$$

Its diagonal contraction is Blaha et al., Eq. (19). Appendix G, Eq. (G1), provides the expanded LAPW expression. A complete implementation includes the muffin-tin term, the interstitial term, the APW boundary/surface term, and the selected scalar-relativistic replacement.

5.3 The energy-linear correction kernel

The WIEN2k routine `VAL_CORR_STRESS` evaluates a one-sided quadratic kernel that is linear in a supplied scalar energy e,

$$K^{\nu}(e) = h^{\nu} - e\, q^{\nu}. \tag{5.11}$$

Here h^{ν} and q^{ν} denote the routine's affine-strain kernels. Their sign relative to Eqs. (5.4)–(5.6) is fixed by the WIEN2k generator convention and by the DFT reduction. The source accumulates

$$D^{\mathrm{corr,impl}}_{\nu} = -2\,\mathrm{Re}\,\mathrm{Tr}(\rho h^{\nu}) + 2\,\mathrm{Re}\,\mathrm{Tr}(\rho_E q^{\nu}). \tag{5.12}$$

The two required pieces are extracted without rewriting hundreds of angular terms:

$$K^{\nu}(0) = h^{\nu}, \qquad K^{\nu}(0) - K^{\nu}(1) = q^{\nu}. \tag{5.13}$$

Blaha's fully expanded diagonal correction is Appendix G, Eq. (G2). The off-diagonal generalization is not separately printed in that paper; it is obtained by Hermitian polarization as above.

6 Why pseudo-band diagonalization gives the off-diagonal result

Let A be any Hermitian band-space matrix and diagonalize a Hermitian density matrix,

$$\rho = V f V^\dagger, \qquad f = \mathrm{diag}(f_p). \tag{6.1}$$

Define pseudo-states

$$|\chi_p\rangle = \sum_n V_{np} |\psi_n\rangle. \tag{6.2}$$

Then

$$\mathrm{Tr}(\rho A) = \sum_p f_p \langle \chi_p | A | \chi_p \rangle. \tag{6.3}$$

Equation (6.3) is exact and includes every off-diagonal $m \neq n$ term. It does not assume that ρ is positive; signed eigenvalues are allowed. The code applies this identity twice:

$$\rho = V f V^\dagger \qquad \text{for } T^\nu \text{ and } h^\nu, \tag{6.4}$$

$$\rho_E = W g W^\dagger \qquad \text{for } q^\nu. \tag{6.5}$$

For a pseudo-state with vector v_p,

$$k_p^\nu(0) = v_p^\dagger h^\nu v_p, \tag{6.6}$$

$$k_p^\nu(0) - k_p^\nu(1) = v_p^\dagger q^\nu v_p. \tag{6.7}$$

Therefore the diagonal WIEN2k quadratic kernel acts as a matrix-free implementation of the generalized matrices.

Implemented. The current code does not store dense arrays T^ν_{mn}, $\mathcal{B}^\nu_{mn}$, or $\mathcal{O}^\nu_{mn}$. It implements their Hermitian contractions by pseudo-band polarization in `cmp_dmft_stress.f90`.

7 Exact mapping to the present stress implementation

The present matrix-free path is:

1. Diagonalize ρ and rotate LAPW $A/B/C$ coefficients to its pseudo-band eigenvectors.
2. Evaluate muffin-tin kinetic stress with `val_kinstrmt`.
3. Add the scalar-relativistic replacement with `val_rela_kinstrmt` when `CALC=RELA`.
4. Evaluate the APW kinetic surface term with `stress_APW_surface`.
5. Set the energy argument to zero and call `val_corr_stress`; this gives h^ν.
6. Evaluate interstitial kinetic stress once per spin and **k** point.
7. Diagonalize ρ_E, rotate coefficients to its pseudo-bands, and call `val_corr_stress` at $e = 0$ and $e = 1$; the difference gives q^ν.
8. Rotate equivalent-atom tensors to global Cartesian axes, reduce over MPI ranks, and apply crystal symmetrization.

The non-spherical correction is decomposed into matching, radial, and angular parts. The current conservative default uses the trace combination; a literal full affine form is available as a diagnostic. This choice is part of the DFT stress audit and is not a new DMFT term.

Important qualification. "Implemented off-diagonal matrices" means that the contractions needed by Hermitian ρ and ρ_E are implemented. The code cannot currently print an arbitrary individual matrix element T^{ν}_{mn}, B^{ν}_{mn}, or O^{ν}_{mn} without a new matrix-assembly diagnostic.

8 Raw and normalized correlated projectors

8.1 Definitions

Let the raw finite-window projector be

$$(P_{\mathbf{k}})_{n\ell} = \langle \psi_{n\mathbf{k}} | \varphi_{\ell} \rangle, \qquad P_{\mathbf{k}} \in \mathbb{C}^{N_b \times N_c}. \tag{8.1}$$

The global correlated-orbital overlap and its inverse square root are

$$S = \sum_{\mathbf{k}} w_{\mathbf{k}} P^{\dagger}_{\mathbf{k}} P_{\mathbf{k}}, \qquad X = S^{-1/2}. \tag{8.2}$$

The projector used by the lattice problem is

$$\boxed{U_{\mathbf{k}} = P_{\mathbf{k}} X.} \tag{8.3}$$

It obeys the global normalization

$$\sum_{\mathbf{k}} w_{\mathbf{k}} U^{\dagger}_{\mathbf{k}} U_{\mathbf{k}} = I_c, \tag{8.4}$$

but generally not $U^{\dagger}_{\mathbf{k}} U_{\mathbf{k}} = I_c$ at each $\mathbf{k}$.

8.2 Complete derivative

The fixed weights satisfy $w_{\mathbf{k},\nu} = 0$. Hence

$$U_{\mathbf{k},\nu} = P_{\mathbf{k},\nu} X + P_{\mathbf{k}} X_{,\nu}, \tag{8.5}$$

$$S_{,\nu} = \sum_{\mathbf{k}} w_{\mathbf{k}} \left(P^{\dagger}_{\mathbf{k},\nu} P_{\mathbf{k}} + P^{\dagger}_{\mathbf{k}} P_{\mathbf{k},\nu} \right). \tag{8.6}$$

The derivative of an inverse square root is a matrix Fréchet derivative. If $S = V \,\mathrm{diag}(s_i) V^{\dagger}$, then

$$(V^{\dagger} X_{,\nu} V)_{ij} = f^{[1]}(s_i, s_j)(V^{\dagger} S_{,\nu} V)_{ij}, \qquad f(s) = s^{-1/2}, \tag{8.7}$$

where

$$f^{[1]}(s_i, s_j) = \begin{cases} \dfrac{s_i^{-1/2} - s_j^{-1/2}}{s_i - s_j}, & s_i \neq s_j, \\ -\dfrac{1}{2} s_i^{-3/2}, & s_i = s_j. \end{cases} \tag{8.8}$$

Equivalently, with $Y = S^{1/2} = X^{-1}$,

$$YX_{,\nu} + X_{,\nu}Y = -XS_{,\nu}X. \tag{8.9}$$

Differentiating Eq. (8.4) supplies a stringent check,

$$\sum_{\mathbf{k}} w_{\mathbf{k}} \left(U^{\dagger}_{\mathbf{k},\nu} U_{\mathbf{k}} + U^{\dagger}_{\mathbf{k}} U_{\mathbf{k},\nu} \right) = 0. \tag{8.10}$$

Implemented. The normalization branch $P_{\mathbf{k}} X_{,\nu}$ is present through the relative derivative $X^{-1}X_{,\nu}$ in `AddProjectorStressNormalization`. It is the stress analogue of the normalization correction added to the DFT+eDMFT force.

8.3 What belongs to $P_{\mathbf{k},\nu}$

It is useful to write

$$P_{\mathbf{k},\nu} = P^{\mathrm{geom}}_{\mathbf{k},\nu} + P^{\mathrm{radial}}_{\mathbf{k},\nu} + P^{\mathrm{subspace}}_{\mathbf{k},\nu} + P^{\mathrm{axes}}_{\mathbf{k},\nu}. \tag{8.11}$$

The terms mean:

Geometry. Reciprocal vectors, phase factors, angular functions, matching coefficients, volume normalization, APW and LO contributions.

Radial. A change of the stored radial projector itself. This is zero in the base convention because the same `projectorw.dat`, R_{MT}, and outer radius are used at every strain.

Subspace. Rotation of the fixed band-index window into complementary stored bands; Sec. 11.

Axes. Rotation of a geometry-following correlated frame. It is zero for the fixed-axis base implementation.

For a current spherical muffin-tin domain with fixed R_{MT}, the atom-centered coordinate $\mathbf{x} = \mathbf{r} - \boldsymbol{\tau}_a$ spans the same fixed ball $|\mathbf{x}| < R_{\mathrm{MT}}$ at every strain. Therefore no extra Reynolds boundary term is generated merely because a radial projector is finite at R_{MT}. For projector 5, the tail outside R_{MT} must be included consistently; then R_{MT} is not a support boundary at all.

9 The one-particle projector gradient $C_{\mathbf{k}}$

At fixed $\widetilde{\Sigma}$, the embedded self-energy varies as

$$\delta\Sigma^{\mathrm{emb}}_{\mathbf{k}} = (\delta U_{\mathbf{k}})\widetilde{\Sigma}U^{\dagger}_{\mathbf{k}} + U_{\mathbf{k}}\widetilde{\Sigma}(\delta U_{\mathbf{k}})^{\dagger}. \tag{9.1}$$

Define

$$\boxed{C_{\mathbf{k}} = \frac{w_{\mathbf{k}}}{\beta} \sum_j \widetilde{\Sigma}(i\omega_j) U^{\dagger}_{\mathbf{k}} G_{\mathbf{k}}(i\omega_j)} \quad \in \mathbb{C}^{N_c \times N_b}. \tag{9.2}$$

The explicit normalized-projector derivative is

$$\boxed{D^{\mathrm{proj}}_{\nu} = 2\,\mathrm{Re} \sum_{\mathbf{k}} \mathrm{Tr}_c(C_{\mathbf{k}} U_{\mathbf{k},\nu}).} \tag{9.3}$$

The factor $2\,\mathrm{Re}$ accounts for the two Hermitian-conjugate branches. If both branches are explicitly constructed, it must not be applied a second time.

Using Eq. (4.6),

$$\boxed{U_{\mathbf{k}} C_{\mathbf{k}} = B_{L,\mathbf{k}} = \rho_{E,\mathbf{k}} - \epsilon_{\mathbf{k}} \rho_{\mathbf{k}}.} \tag{9.4}$$

This is a band-space identity. $U_{\mathbf{k}} C_{\mathbf{k}}$ is $N_b \times N_b$, while $C_{\mathbf{k}} U_{\mathbf{k},\nu}$ is $N_c \times N_c$.

10 SVD, pseudoinverse reconstruction, and what can be lost

Let

$$U_{\mathbf{k}} = L_{\mathbf{k}} s_{\mathbf{k}} R_{\mathbf{k}}^{\dagger} \tag{10.1}$$

be a singular-value decomposition. A small orbital-space dimension makes $U^{\dagger}U$ inexpensive to invert, but does not guarantee it is nonsingular. Symmetry or an insufficient band window can make one or more singular values zero at an individual $\mathbf{k}$.

If $U_{\mathbf{k}}$ has full column rank,

$$C_{\mathbf{k}} = (U_{\mathbf{k}}^{\dagger} U_{\mathbf{k}})^{-1} U_{\mathbf{k}}^{\dagger} B_{L,\mathbf{k}}. \tag{10.2}$$

The implemented general reconstruction uses the Moore–Penrose pseudoinverse,

$$\boxed{C_{\mathbf{k}}^{\mathrm{MP}} = U_{\mathbf{k}}^{\mathrm{MP}} B_{L,\mathbf{k}} = (U_{\mathbf{k}}^{\dagger} U_{\mathbf{k}})^{\mathrm{MP}} U_{\mathbf{k}}^{\dagger} B_{L,\mathbf{k}}.} \tag{10.3}$$

The notation A^{MP} is used to avoid confusing a pseudoinverse with a Hermitian conjugate $A^{\dagger}$.

If Eq. (9.4) is exact for a directly accumulated C^{dir}, then

$$C_{\mathbf{k}}^{\mathrm{MP}} = U_{\mathbf{k}}^{\mathrm{MP}} U_{\mathbf{k}} C_{\mathbf{k}}^{\mathrm{dir}} \equiv \Pi_{\mathbf{k}} C_{\mathbf{k}}^{\mathrm{dir}}, \tag{10.4}$$

where $\Pi_{\mathbf{k}}$ projects onto the local subspace represented by the chosen band window. Hence

$$C_{\mathbf{k}}^{\mathrm{dir}} = C_{\mathbf{k}}^{\mathrm{MP}} + C_{\mathbf{k}}^{\perp}, \tag{10.5}$$

$$U_{\mathbf{k}} C_{\mathbf{k}}^{\perp} = 0. \tag{10.6}$$

Although $C^{\perp}$ is invisible in $B_L = UC$, it can contribute to the stress:

$$\Delta D_{\nu}^{\mathrm{proj}} = 2\,\mathrm{Re}\,\mathrm{Tr}(C_{\mathbf{k}}^{\perp} U_{\mathbf{k},\nu}). \tag{10.7}$$

This is the precise reason the equation-of-motion reconstruction can be insufficient at a rank-deficient $\mathbf{k}$ point.

Implemented. The current `BuildWindowResponseMatrices` diagonalizes the small Gram matrix $U_W^{\dagger} U_W$, applies a relative eigenvalue cutoff, and constructs Eq. (10.3). It reports the reconstruction residual when projector diagnostics are enabled.

11 Fixed-index window and complementary-band response

The DMFT window membership is fixed by band index, so there is no derivative of a sharp energy cutoff. The selected eigenvectors nevertheless rotate into bands outside the window. For a nondegenerate window state n,

$$|\psi_{n,\nu}\rangle_\perp = \sum_{a\notin W} |\psi_a\rangle \frac{\langle\psi_a|H_{,\nu} - \epsilon_n O_{,\nu}|\psi_n\rangle}{\epsilon_n - \epsilon_a}. \tag{11.1}$$

The exact sum contains all complementary bands. The implementation uses the additional bands already present in `case.vector` or `case.vectorso`; convergence can be improved by asking LAPW1 to write more bands.

Let U_{full} contain projector rows for every stored band and let

$$K_{\text{full}} = U_{\text{full}} C^{\text{MP}}. \tag{11.2}$$

For $a \notin W$ and $n \in W$, the code builds auxiliary Hermitian response matrices

$$(\rho^{\text{resp}})_{an} = \frac{(K_{\text{full}})_{an}}{\epsilon_n - \epsilon_a}, \qquad (\rho^{\text{resp}})_{na} = (\rho^{\text{resp}})^*_{an}, \tag{11.3}$$

$$(\rho^{\text{resp}}_E)_{an} = \epsilon_n (\rho^{\text{resp}})_{an}, \qquad (\rho^{\text{resp}}_E)_{na} = (\rho^{\text{resp}}_E)^*_{an}. \tag{11.4}$$

The corresponding response is evaluated by reusing the LAPW kernels,

$$D^{\text{resp}}_\nu = \text{Tr}(\rho^{\text{resp}} H_{,\nu}) - \text{Tr}(\rho^{\text{resp}}_E O_{,\nu}), \tag{11.5}$$

with the source's calibrated sign convention. The response matrices are a contraction device; they must not be added to the physical charge density. Near cross-window degeneracies, ordinary denominators are not reliable and a degenerate spectral-projector treatment is ultimately preferable.

12 Direct $C_{\mathbf{k}}$ accumulation: proposed comparison switch

The pseudoinverse can be eliminated by accumulating Eq. (9.2) inside the existing Matsubara loop. At a positive Matsubara frequency the paired contribution is

$$\widetilde{\Sigma}(i\omega) U^\dagger G(i\omega) + \widetilde{\Sigma}^\dagger(i\omega) U^\dagger G^\dagger(i\omega). \tag{12.1}$$

Because C is rectangular and generally non-Hermitian, it must not be Hermitian-symmetrized. The negative-frequency partner in Eq. (12.1) is not $C(i\omega)^\dagger$; that object has transposed dimensions and a different matrix order.

Proposed diagnostic / not yet implemented. Add `EDMFT_STRESS_C_MODE=PINV|DIRECT|COMPARE`. `PINV` retains the current production path. `DIRECT` uses Eq. (9.2). `COMPARE` computes both while initially leaving the production result unchanged.

The comparison should report

$$r_C = \frac{\left\| C^{\text{dir}} - C^{\text{MP}} \right\|_F}{\max(\|C^{\text{dir}}\|_F, \epsilon_{\text{num}})}, \tag{12.2}$$

$$r_B^{\mathrm{dir}} = \frac{\left\| UC^{\mathrm{dir}} - B_L \right\|_F}{\max(\|B_L\|_F, \epsilon_{\mathrm{num}})}, \tag{12.3}$$

$$r_B^{\mathrm{MP}} = \frac{\left\| UC^{\mathrm{MP}} - B_L \right\|_F}{\max(\|B_L\|_F, \epsilon_{\mathrm{num}})}, \tag{12.4}$$

$$\Delta\sigma_\nu^{\mathrm{proj}} = \frac{2}{\Omega} \operatorname{Re} \operatorname{Tr} \left[(C^{\mathrm{dir}} - C^{\mathrm{MP}}) U_{,\nu} \right]. \tag{12.5}$$

Three band-space objects should also be compared independently:

1. $B_L^{\mathrm{EOM}} = \rho_E - \epsilon\rho$;
2. $B_L^{\mathrm{band}} = T \sum \Sigma^{\mathrm{emb}} G$, already accumulated by a diagnostic in `cmp_dmft_weights2`;
3. $B_L^{\mathrm{local}} = UC^{\mathrm{dir}}$.

This three-way test separates an energy-moment/tail problem from a local projector mapping problem and from information lost by the pseudoinverse.

13 Matsubara summation used by ρ_E and the proposed direct C

The implementation uses two frequency regions:

1. the first consecutive Matsubara frequencies are summed explicitly;
2. the high-frequency region is stored on a sparse logarithmic mesh and summed with precomputed cardinal cubic-spline weights `matsum_weight`.

The nonlinear resolvent is formed first; the final matrix-valued integrand is then multiplied by the cubic-spline quadrature weight. This is more accurate than interpolating poles or eigenvalues before forming the resolvent.

For the dynamic self-energy tail,

$$\widetilde{\Sigma}(i\omega) - \Sigma_\infty = \frac{\Sigma_1}{i\omega} + \frac{\Sigma_2}{(i\omega)^2} + \cdots. \tag{13.1}$$

The code adaptively estimates Σ_1: a smooth tail uses a two-moment fit, a noisy tail uses a short plateau average, and a clearly bad or increasing tail falls back to the robust endpoint estimate. In the energy-moment sum, the leading $\Sigma_1/(i\omega)^2$ contribution is subtracted from the sparse cubic quadrature and restored over exactly the same frequency interval.

If the positive-frequency sparse tail starts at fermionic index j_0 with $\omega_j = (2j-1)\pi/\beta$, the scalar used by the code is

$$M_2^+(j_0) = \sum_{j=j_0}^{\infty} \frac{1}{\omega_j^2} = \frac{\beta^2}{8} - \sum_{j=1}^{j_0-1} \left[\frac{\beta}{(2j-1)\pi} \right]^2. \tag{13.2}$$

The final paired Matsubara factor supplies $2/\beta$. Equation (13.2) is safe because it is applied only to the high-frequency tail after the same term has been removed from the numerical residual; it is not added as a singular approximation at low frequency.

The direct-C accumulator must use exactly the same cubic weights, positive and negative matrix ordering, Σ_1 subtraction, and analytic restoration as B_L. Otherwise $UC^{\mathrm{dir}} = B_L$ would fail for a purely numerical reason.

14 Standard DFT stress channels retained in eDMFT

The electronic DMFT generalization does not replace the remaining DFT stress machinery. The full result retains:

- valence kinetic stress, including muffin-tin, interstitial, APW surface, and relativistic corrections;
- valence incomplete-basis/Pulay correction;
- core correction from `lcore`;
- electrostatic and nuclear contributions from `lapw0`;
- exchange-correlation stress, including the GGA contribution where applicable;
- spin-channel assembly and final crystal symmetrization;
- the explicit normalized-projector term and fixed-window response.

The eDMFT matrices ρ and ρ_E replace diagonal DFT occupations only inside the valence one-particle contractions. A naive substitution of diagonal occupations loses off-diagonal coherence and gives the overlap term the wrong energy weight.

15 Source-level equation map

Mathematical object	Current source location	Implementation strategy
ρ, ρ_E	`cmp_dmft_weights.f90`	Matsubara sum, cubic sparse-tail quadrature, analytic Σ_1 restoration.
T^ν contraction	`cmp_dmft_stress.f90`	ρ pseudo-bands passed to muffin-tin, interstitial, APW surface, and RELA kernels.
h^ν contraction	`cmp_dmft_stress.f90`	`val_corr_stress` evaluated at $e = 0$.
q^ν contraction	`cmp_dmft_stress.f90`	ρ_E pseudo-bands; $K(0) - K(1)$.
$U_{,\nu}$ geometry	`l2main.f90`, projector strain routines	Fixed-coefficient raw-projector derivative.
$PX_{,\nu}$	`AddProjectorStress-Normalization`	Relative inverse-square-root derivative.
C^{MP}	`BuildWindowResponse-Matrices`	Eigenvalue-cutoff Moore–Penrose inverse of $U_W^\dagger U_W$.
Complementary bands	`BuildWindowResponse-Matrices`	Eqs. (11.4)–(11.5) using stored LAPW1 bands.
$T \sum \Sigma^{\mathrm{emb}} G$ check	`cmp_dmft_weights.f90`	Direct band-space diagnostic with the same Matsubara tail.
C^{dir}	not yet present	Proposed local-by-band accumulation, Eq. (9.2).

16 What has been established and what remains open

16.1 Established algebraic facts

- The first derivative requires no impurity two-particle vertex.
- ρ_E, not $\epsilon\rho$, multiplies the generalized-overlap derivative.
- Blaha's diagonal Eqs. (19) and (20) generalize uniquely to the Hermitian bilinear matrices in Sec. 5.
- Pseudo-band diagonalization evaluates the required off-diagonal contractions exactly for Hermitian

ρ and ρ_E.
- The projector must be differentiated after normalization: $U_{,\nu} = P_{,\nu}X + PX_{,\nu}$.
- A fixed band-index window still has a complementary-band eigenvector response.
- C^{MP} can discard a null-space component that contributes through $U_{,\nu}$.

16.2 Current numerical audit state

Direct finite differences of complete LAPW H and O matrices satisfy the generalized eigenvalue checks and reproduce finite differences of DFT band energies. This validates the operator-level paired expression $\mathrm{Tr}(\rho H_{,\nu}) - \mathrm{Tr}(\rho_E O_{,\nu})$. The principal unresolved problems are:

1. the current analytic isolated $M_\Sigma O_{,\nu}$ channel does not equal the complete metric derivative by one universal sign or factor;
2. DFT baseline residuals are not yet uniformly below the desired sub- 0.1 GPa scale in every material;
3. the effect of $C^\perp$ at singular $\mathbf{k}$ points has not been measured because C^{dir} is not yet accumulated;
4. RELA operator-level closure, shear components, and magnetic two-spin assembly remain later validation stages.

The recommended nonmagnetic comparison sequence is $SrVO_3$, MnO, FeSe NREL, and FeSe RELA; magnetic MnTe should follow after the nonmagnetic identities close.

17 Validation protocol

17.1 Algebraic checks at every k

$$\delta_O = \left\| A^\dagger O A - I \right\|_F , \tag{17.1}$$

$$\delta_H = \left\| A^\dagger H A - \epsilon \right\|_F , \tag{17.2}$$

$$\delta_E = \left\| \rho_E - \tfrac{1}{2}\{\epsilon, \rho\} - M_\Sigma \right\|_F , \tag{17.3}$$

$$\delta_N = \left\| \sum_{\mathbf{k}} w_{\mathbf{k}} (U^\dagger_{,\nu} U + U^\dagger U_{,\nu}) \right\|_F , \tag{17.4}$$

$$\delta_C = \left\| UC - B_L \right\|_F . \tag{17.5}$$

The singular spectrum of $U^\dagger_{\mathbf{k}} U_{\mathbf{k}}$ should be printed together with all pseudoinverse cutoffs.

17.2 Finite-difference checks

For a central step h,

$$D^{\mathrm{FD}}_\nu(h) = \frac{\Gamma(+h) - \Gamma(-h)}{2h}. \tag{17.6}$$

Use at least two steps and, where smooth, Richardson extrapolation,

$$D^{\mathrm{Rich}}_\nu = \frac{4D^{\mathrm{FD}}_\nu(h/2) - D^{\mathrm{FD}}_\nu(h)}{3}. \tag{17.7}$$

The $-h$, 0, and $+h$ calculations must have identical R_{MT}, projector radial data, band-index window, number of stored bands, $\mathbf{k}$ mesh, and reciprocal cutoffs. At frozen Σ, the DFT charge must be relaxed. For a fully stationary Klein comparison the impurity functional difference $\Phi_{\mathrm{imp}} - \Phi_{\mathrm{DC}}$ must be included consistently; its omission is not an analytic basis error.

18 Compact formula sheet

$$\sigma_\nu = \Omega^{-1}\partial_\nu \Gamma, \tag{18.1}$$

$$G_{\mathbf{k}}^{-1} = (i\omega + \mu)I - \epsilon_{\mathbf{k}} - U_{\mathbf{k}}\widetilde{\Sigma}U_{\mathbf{k}}^\dagger, \tag{18.2}$$

$$\rho_{\mathbf{k}} = T\sum_\omega G_{\mathbf{k}}, \qquad \rho_{E,\mathbf{k}} = T\sum_\omega (i\omega+\mu)G_{\mathbf{k}}, \tag{18.3}$$

$$M_{\Sigma,\mathbf{k}} = \rho_{E,\mathbf{k}} - \tfrac{1}{2}\{\epsilon_{\mathbf{k}}, \rho_{\mathbf{k}}\}, \tag{18.4}$$

$$P_{\mathbf{k}} : N_b \times N_c, \qquad S = \sum_{\mathbf{k}} w_{\mathbf{k}} P_{\mathbf{k}}^\dagger P_{\mathbf{k}}, \tag{18.5}$$

$$X = S^{-1/2}, \qquad U_{\mathbf{k}} = P_{\mathbf{k}} X, \tag{18.6}$$

$$U_{\mathbf{k},\nu} = P_{\mathbf{k},\nu}X + P_{\mathbf{k}}X_{,\nu}, \tag{18.7}$$

$$C_{\mathbf{k}} = \frac{w_{\mathbf{k}}}{\beta}\sum_\omega \widetilde{\Sigma}U_{\mathbf{k}}^\dagger G_{\mathbf{k}}, \qquad D_\nu^{\mathrm{proj}} = 2\,\mathrm{Re}\sum_{\mathbf{k}} \mathrm{Tr}(C_{\mathbf{k}}U_{\mathbf{k},\nu}), \tag{18.8}$$

$$B_{L,\mathbf{k}} = \rho_{E,\mathbf{k}} - \epsilon_{\mathbf{k}}\rho_{\mathbf{k}} = U_{\mathbf{k}}C_{\mathbf{k}}, \tag{18.9}$$

$$C_{\mathbf{k}}^{\mathrm{MP}} = (U_{\mathbf{k}}^\dagger U_{\mathbf{k}})^{\mathrm{MP}} U_{\mathbf{k}}^\dagger B_{L,\mathbf{k}}, \tag{18.10}$$

$$T_{mn}^{\alpha\beta} = \tfrac{1}{2}\langle\psi_m|\partial_\alpha\partial_\beta + partial_\beta\partial_\alpha|\psi_n\rangle, \tag{18.11}$$

$$\mathcal{B}^\nu = b^\nu + b^{\nu\dagger}, \qquad \mathcal{O}^\nu = o^\nu + o^{\nu\dagger}, \tag{18.12}$$

$$b_{mn}^\nu = \langle\psi_m^{B,\nu}|H|\psi_n\rangle, \qquad o_{mn}^\nu = \langle\psi_m^{B,\nu}|\psi_n\rangle, \tag{18.13}$$

$$K^\nu(e) = h^\nu - eq^\nu, \qquad q^\nu = K^\nu(0) - K^\nu(1). \tag{18.14}$$

19 Recommended next implementation step

The next source change should be narrowly diagnostic:

1. accumulate $C_{\mathbf{k}}^{\mathrm{dir}}$ inside `cmp_dmft_weights2`, reusing the existing G and Σ at no additional matrix inversion cost;
2. apply the identical cubic Matsubara weights and high-frequency tail restoration used for B_L;
3. add `PINV`, `DIRECT`, and `COMPARE` modes;
4. in `COMPARE`, print Eqs. (12.5) and both projector stress tensors without altering the default production answer;
5. run the same executable on $SrVO_3$, MnO, FeSe NREL/RELA, then MnTe.

If C^{dir} and C^{MP} agree in stress despite singular values, the remaining error lies in the LAPW H/O kernels or another stress channel. If they differ by the observed numerical residual, Eq. (10.7) identifies the missing projector-gradient component directly.

A Derivation of the inverse-square-root Sylvester equation

The normalized metric satisfies

$$XSX = I. \tag{A.1}$$

Differentiating gives

$$X_{,\nu}SX + XS_{,\nu}X + XSX_{,\nu} = 0. \tag{A.2}$$

Since $SX = XS = S^{1/2} = Y$,

$$X_{,\nu}Y + YX_{,\nu} = -XS_{,\nu}X, \tag{A.3}$$

which is Eq. (8.9). The solution is unique when S is positive definite. If the global S itself is singular, the requested correlated set is not represented by the total selected band manifold and an SVD/truncated-subspace definition is required before differentiating.

B Polarization proof for the generalized kernels

For $\rho = VfV^\dagger$,

$$\mathrm{Tr}(\rho A) = \mathrm{Tr}(VfV^\dagger A) = \mathrm{Tr}(fV^\dagger AV) \tag{B.1}$$

$$= \sum_p f_p (V^\dagger AV)_{pp} = \sum_p f_p v_p^\dagger A v_p. \tag{B.2}$$

Thus a routine capable of evaluating $v^\dagger Av$ for an arbitrary linear combination of Kohn–Sham states evaluates the full off-diagonal contraction. The same argument applies independently to ρ_E, even when its eigenvalues are negative.

C Relation to Blaha's numbered equations

Blaha equation	Published object	Present generalization
Eq. (19)	Diagonal valence kinetic stress $T_{nn}^{\alpha\beta}$	Hermitian matrix $T_{mn}^{\alpha\beta}$, Eq. (5.10), contracted with off-diagonal ρ.
Eq. (20)	Diagonal IBS/Pulay form $2\,\mathrm{Re}\langle\psi_n^{B,\nu}\vert H-\epsilon_n\vert\psi_n\rangle$	Matrices $\mathcal{B}^\nu$ and $\mathcal{O}^\nu$, Eqs. (5.4)–(5.9).
Eq. (G1)	Expanded diagonal kinetic LAPW formula	Reused on ρ pseudo-bands.
Eq. (G2)	Expanded diagonal valence-correction formula	Reused on ρ and ρ_E pseudo-bands through $K(0)$ and $K(0)-K(1)$.

Blaha's paper does not list separate $m \neq n$ matrix elements. Their Hermitian polarization is required only when the one-particle density matrices are off-diagonal, as in DFT+eDMFT.

D References and source provenance

1. K. Belbase, A. Tröster, and P. Blaha, "Stress tensor in the linearized augmented plane wave method," *Phys. Rev. B* **104**, 174113 (2021).
2. K. Haule, *Forces in the LAPW implementation of DFT+eDMFT*, internal derivation in `src/doc/Forces_LAPW.pdf`.
3. Current implementation inspected in `eDMFT_new/src/dmft2/cmp_dmft_stress.f90`, `eDMFT_new/src/dmft2/cmp_dmft_weights.f90`, `eDMFT_new/src/dmft2/l2main.f90`, and `eDMFT_new/src/dmft2/l2mainSubs.f90`.
4. Current audit summary: `eDMFT_new/src/doc/edmft_stress_audit_state_20261105.md`.

End of technical working note. No production source code was modified in its preparation.